

Candidate Number _____

Fraction _____

2017 International Biology Olympiad National Team Selection Camp Practical Test Questions

The **third** exam session included practical questions on plant structure and physiology.

Note: Operation time is limited, please make good use of the time waiting for the experiment to take effect.

Materials and medicines:

A. Experimental Plant		quantity
Materials: 1. Apple, guava, and kumquat		Half each
fruits 2. Apple, guava, and kumquat juice		1 tube each
3. Apple, guava, and kumquat B.		1 gram each
Experimental	Quantity B: Experimental Equipment:	quantity
Equipment:	Shared acid-base test paper	1 box
Centrifuge,	box; Shared micropipettes (P1000, 3	1 piece
spectrophotometer,	sets); P200 micropipettes, 1	1 piece
mortar and	micropipette tip (1000 µl); 1 micropipette	1 box
pestle,	tip (20–200 µl); 8 colorimetric tubes; 8 optical	1 box
fruit	microscopes; 1 single-	8
knife, cutting board,	edged blade; 1 glass slide/	1 unit
tweezers, 1.5 ml centrifuge	coverslip; 8 sheets of	1 piece
tubes (brown),	A4 paper; stapler; oil-based pen.	10 groups
1.5 ml	Quantity C:	1 sheet
centrifuge tubes, computer, waste liquid cup, labels, stickers,	Chemical Solution:	1 piece
	25 ml DPPH	1 piece
C. Drug solution: 1.	methanol solution.	quantity
80% methanol		2 ml

1. Please confirm that your candidate number is correct; if it is incorrect, please raise your hand and ask the teaching assistant for assistance.
2. The medicines and equipment on the table will not be replenished after use. Please use the shared equipment according to the instructions.
3. This exam paper (including the cover and the exam paper) has a total of 8 pages and must be returned in its entirety when submitting the exam.
4. The time allotted for answering the questions is 90 minutes. Please answer the questions on this paper.
5. Answers may be written on the back of the questions, but please indicate the question number.

Plant Structure and Physiology Practical Problems

Analysis of the fruit's post-harvest metabolites, pH, and antioxidant properties

I. Note

This practical exercise includes observation of plant structure and testing of physiological functions, each accounting for 50%. Please browse each section first.

Practical instructions are provided to help you plan your time effectively so that you can successfully complete all the practical projects.

II. Practical Principles

After a plant undergoes metabolism, some of its metabolic byproducts gradually accumulate in the cells, forming a substance called [a substance called a plant's metabolic byproduct].

Post-biomass, such as starch granules, oil droplets, crystals, etc.

The metabolic byproducts of plant cells also include free radicals, which can cause oxidative damage to cells.

Antioxidants in cells can effectively scavenge free radicals. This experiment aims to examine the ability of samples to scavenge 2,2-diphenyl-1-picrylhydrazyl (DPPH) free radicals and to understand the antioxidant capacity of plant tissues. Current research confirms that aging is related to oxidative damage to biomolecules caused by free radicals, which are molecules containing unpaired electrons.

These chemical species can steal electrons from other molecules, causing them to oxidize. Most free radicals are unstable and reactive.

Intermediates readily undergo chemical reactions. Only a very few free radicals are stable, such as DPPH, because...

This is widely used in assessing the antioxidant capacity of substances. The DPPH methanol solution used in this experiment is purple.

It exhibits strong absorbance at 517 nm. If it binds to the sample, the absorbance will decrease, thus allowing for the determination of the sample's absorbance.

The lower the absorbance value, the greater the ability of the sample to scavenge DPPH free radicals.

The stronger the force.

III. Experimental Methods

(I) Fruit structure and metaplastic substances in cells (the table below is a reference for relevant terms)

1. Use a blade to cut open the fruit, observe and draw a diagram to indicate the fruit's structure.

2. Take samples from different parts of different fruits, examine the tissue cells under a microscope, identify the tissues containing different metaplastic substances, draw diagrams of the observed tissue cells and metaplastic substances, and select the three best slides from the prepared specimens. Label each slide with the corresponding number from the diagram on the answer sheet. After the practical exercise, these three slides must be submitted for checking and grading; failure to submit slides will result in a deduction of 3 points for this question's drawing.

Related terms: fruit structure

(exocarp, mesocarp,	Post-biomass starch
endocarp, placenta, seed, juice sac, receptacle)	grains, oil droplets, oil cavities, needle-like crystals, clustered crystals (such as spiky spheres), stalactite-like crystals, polyhedral crystals, and crystal sand.

(II) Analysis of Fruit pH and Antioxidant Capacity

1. Using the juice from the three prepared fruits, tear off a small piece of acid-base test paper, about 2 cm long, with tweezers.

After soaking in fruit juice, remove and place on A4 paper. Observe the color blocks in the acid-base test paper kit to measure the pH value.

Please record the completion on page 7.

2. Using one gram each of apple and guava (weighed) and one gram of kumquat juice (squeezed into a small centrifuge tube)...

(The contents of the container) were ground into homogenate in a mortar with 3 ml of 80% methanol, and then 1 ml was taken with a micropipette and...

Give the teaching assistant to help with centrifugation for 10 minutes.

Please repeat the following experimental steps twice: Please conduct both sets of experiments simultaneously.

3. Please use brown centrifuge tubes to take 20 μ l of the supernatant of the fruit juice and 20 μ l of 80% methanol (this tube is the control group).

And label them with stickers.

4. Add 80% methanol to a final volume of 1 ml.

5. Add 200 μ l of DPPH methanol solution to each, mix well, let stand for 15 minutes, and then apply a microparticle of P1000.

Use a pipette to transfer the solution to the colorimetric tube.

6. Please remember the order of colorimetric tube administration, and then have the teaching assistant assist you in measuring the absorbance at 517 nm on the spectrophotometer.

Light value.

7. Staple the spectrophotometer measurement results to page 7.

8. The control group is DPPH methanol solution.

DPPH free radical scavenging capacity % = [(control group absorbance at 517 nm - sample absorbance at 517 nm)]

[Absorbance value / Absorbance value of control group at 517 nm] x 100

V. Experimental Results and Discussion:

(I) Fruit Structure and Metabolites in Cells 1. Draw a

diagram of the fruit structure and label the names of each part. (15%)

Fruit name:
Fruit name:
Fruit name:

1. Draw diagrams of the tissue cells and metaplastic material you observe (draw up to 6 types). Select the three best slides. _____

Return the drawings for inspection, and affix labels to each drawing, indicating the corresponding number on the answer sheet. (35%)

Note: The coordinates of the target object read from the microscope stage = (X-axis graduations; Y-axis graduations)

Fruit name: Part of tissue sampled: Metabolite:

Submit slide number: See the coordinates of the biomass after viewing: (_____ ; _____)

Fruit name: Tissue sampling location: Submitted slide number: Coordinates of the

Post-biomass:

biomass after observation: (_____ ; _____)

Fruit name: Tissue sampling location: Submitted slide number: Coordinates of the

Post-biomass:

biomass after observation: (_____ ; _____)

Fruit name: Tissue sampling location: Submitted slide number: Coordinates of the biomass after observation: (_____)	Post-biomass:
Fruit name: Tissue sampling location: Submitted slide number: Coordinates of the biomass after observation: (_____)	Post-biomass:
Fruit name: Tissue sampling location: Submitted slide number: Coordinates of the biomass after observation: (_____)	Post-biomass:

(II) Analysis of Fruit pH and Antioxidant Capacity

1. What are the pH values of various fruit juices? (10%)
2. What are the spectrophotometer readings and free radical scavenging capacity (%) for various fruit juices? (30%)

Please fill in the answers to questions 1 and 2 in the table below.

	pH value	517nm absorbance		Free radical scavenging ability	
		Test-1	Test-2	Test-1	Test-2
control group					
apple					
Kumquat					
Guava					

Please staple the spectrophotometer results below:

3. Given that the vitamin C content of these three fruits is (guava **7.93**, **kumquat 1.20**, **apple 0**), their acidity is...

Are alkalinity or free radical scavenging ability related to the vitamin C content, respectively? Please explain.

The reason is (10%)

A: pH level —

Free radical scavenging ability —